

PalmSource®

Conference and Expo



Security for the ARM-based OS

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Introduction to the Cryptographic Provider Manager API



Agenda

Fundamental Paradigms

High Level API Overview

Best Practices and Examples



Fundamental Paradigm

- Very simple API for non-cryptographers
- All cryptographic “classes” of operations expose the same functionality
- Applications using defaults will get “reasonable” behavior



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Standard Library APIs

- CPMLibOpen
- CPMLibClose
- CPMLibSleep
- CPMLibWake



Library Management APIs

- CPMLibGetInfo
- CPMLibSetDebugLevel
- CPMLibEnumerateProviders
- CPMLibGetProviderInfo
- CPMLibSetDefaultProvider



RNG APIs

- CPMLibAddRandomSeed
- CPMLibGenerateRandomBytes



Key APIs

Symmetric Keys

- CPMLibGenerateKey
- CPMLibImportKeyInfo
- CPMLibExportKeyInfo
- CPMLibReleaseKeyInfo



Key APIs (contd.)

Asymmetric Keys

- CPMLibGenerateKeyPair
- CPMLibImportKeyPairInfo
- CPMLibExportKeyPairInfo
- CPMLibReleaseKeyPairInfo



Key APIs (contd.)

Derived Keys (shared secret, PBE)

- CPMLibDeriveKeyData
- CPMLibImportKeyDataInfo
- CPMLibExportKeyDataInfo
- CPMLibReleaseKeyDataInfo



Message Digest APIs

- CPMLibHashInit
- CPMLibHashUpdate
- CPMLibHashFinal
- CPMLibHash
- CPMLibImportHashInfo
- CPMLibExportHashInfo
- CPMLibReleaseHashInfo



Cipher APIs

- CPMLib(Encrypt/Decrypt)Init
- CPMLib(Encrypt/Decrypt)Update
- CPMLib(Encrypt/Decrypt)Final
- CPMLib(Encrypt/Decrypt)
- CPMLibImportCipherInfo
- CPMLibExportCipherInfo
- CPMLibReleaseCipherInfo



Public/Private Key Operation APIs

- CPMLib(Sign/Verify)Init
- CPMLib(Sign/Verify)Update
- CPMLib(Sign/Verify)Final
- CPMLib(Sign/Verify)
- CPMLibImport(Sign/Verify)Info
- CPMLibExport(Sign/Verify)Info
- CPMLibRelease(Sign/Verify)Info



More Information on Cryptography

- Applied Cryptography – Bruce Schneier
  - http://www.counterpane.com/applied.html
- Cryptography FAQ
  - http://www.faqs.org/faqs/cryptography-faq/
- RSA Cryptography FAQ
  - http://www.rsasecurity.com/rsalabs/faq/



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Best practices

- For Password Based Encryption do not save the secret anywhere (except user's head)
- For password authentication don't save the password, save the hash of the password
- Do not use the same RC4 key twice
- Use long enough keys (64 – 128 bit)
  - 40 bit keys can be broken in 3 hours



Examples

- See tech notes and URL references at www.palmsource.com
  - Open library example
  - Key derivation example
  - Simple hash example
  - Import/Export of keys and cipher contexts example
  - Simple encrypt/decrypt example



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